



# DUNMAN SECONDARY SCHOOL

CANDIDATE  
NAME

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CLASS

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INDEX  
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## PRELIMINARY EXAMINATION 2024 SECONDARY 4 EXPRESS/ 5 NORMAL ACADEMIC

### MATHEMATICS

Paper 1

**4052/01**

19 August 2024

2 hours 15 minutes

Candidates answer on the Question Paper.

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#### READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

| Calculator Model |
|------------------|
|                  |

Answer **all** questions.

If working is needed for any question it must be shown with the answer.

Omission of essential working will result in loss of marks.

The use of an approved scientific calculator is expected, where appropriate.

If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For  $\pi$ , use either your calculator or 3.142, unless the question requires the answer in terms of  $\pi$ .

The number of marks is given in brackets [ ] at the end of each question or part question.

The total number of marks for this paper is 90.

| Total Marks |
|-------------|
|             |

***Mathematical Formulae****Compound interest*

$$\text{Total Amount} = P \left( 1 + \frac{r}{100} \right)^n$$

*Mensuration*

$$\text{Curved surface area of a cone} = \pi r l$$

$$\text{Surface area of a sphere} = 4\pi r^2$$

$$\text{Volume of a cone} = \frac{1}{3} \pi r^2 h$$

$$\text{Volume of a sphere} = \frac{4}{3} \pi r^3$$

$$\text{Area of triangle } ABC = \frac{1}{2} ab \sin C$$

$$\text{Arc length} = r\theta, \text{ where } \theta \text{ is in radians}$$

$$\text{Sector area} = \frac{1}{2} r^2 \theta, \text{ where } \theta \text{ is in radians}$$

*Trigonometry*

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

*Statistics*

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

$$\text{Standard deviation} = \sqrt{\frac{\sum fx^2}{\sum f} - \left( \frac{\sum fx}{\sum f} \right)^2}$$

- 1 (a) Write  $27^4$  as a power of 3.

*Answer* ..... [1]

(b)  $5^x = 5^6 + 5^6 + 5^6 + 5^6 + 5^6$

Find the value of  $x$ .

*Answer*  $x =$  ..... [1]

- 
- 2 A box measures 54 cm by 78 cm by 204 cm.  
The box is completely filled with identical cubes.

Find the minimum number of cubes required.

*Answer* ..... [2]

- 3 Three integers,  $X$ ,  $Y$  and 175 000, can be written as products of prime factors.

$$X = a^3 \times b^{c+3} \times 7$$

$$Y = a^2 \times b^{c+1} \times 7$$

$$175\,000 = 2^3 \times 5^5 \times 7$$

The lowest common multiple (LCM) of  $X$  and  $Y$  is 175 000.

Write down the values of  $a$ ,  $b$  and  $c$ .

*Answer*  $a = \dots\dots\dots$

$b = \dots\dots\dots$

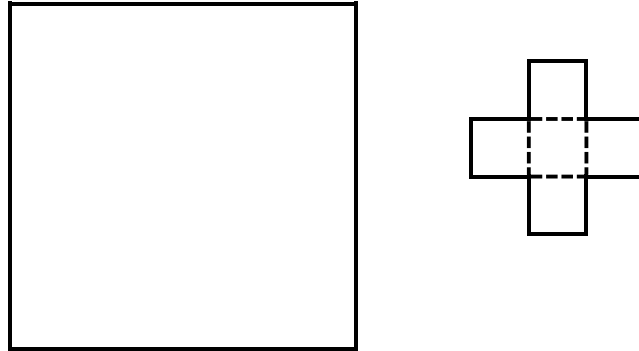
$c = \dots\dots\dots$  [3]

- 4 It is given that  $a = \frac{4+3c^2}{c^2-2b}$ .

Express  $c$  in terms of  $a$  and  $b$ .

Answer  $c = \dots\dots\dots$  [3]

5



The diagram shows a large square and a cross made of five small squares.  
The ratio of the perimeters large square : cross = 2 : 1.

Find the ratio of the areas large square : cross.

Answer  $\dots\dots\dots : \dots\dots\dots$  [2]

6 Solve the equation  $\frac{4x}{3} - \frac{5x-1}{2} = -3$ .

*Answer*  $x = \dots\dots\dots$  [3]

- 7 Tammy invests \$5000 in a savings account at a rate of 4% per year compound interest. After 7 years, she withdraws \$3000 and leaves the remaining amount of the money in the account, which continues to earn interest at the same rate.

Tammy then claims that in the next 7 years, the amount of interest earned will be more than that earned in the first 7 years.

Is Tammy correct?

Show your working to support your answer.

*Answer*

- 8 (a) Simplify  $7(3y + 2x) - 2(x - 4y)$ .

*Answer* ..... [2]

- (b) Factorise completely  $12ax - 6xy - 4by + 8ab$ .

*Answer* ..... [2]

- 9 The heights, in cm, of 25 boys and 28 girls were recorded.  
The results are shown in the stem-and-leaf diagram.

| Boys |   |   |   |   |   |  |    |  |   | Girls |   |   |   |   |  |  |  |  |  |
|------|---|---|---|---|---|--|----|--|---|-------|---|---|---|---|--|--|--|--|--|
|      |   |   |   | 6 | 1 |  | 8  |  | 0 | 2     |   |   |   |   |  |  |  |  |  |
|      | 8 | 5 | 3 | 0 | 0 |  | 9  |  | 1 | 4     | 6 | 8 |   |   |  |  |  |  |  |
|      | 9 | 6 | 5 | 1 | 0 |  | 10 |  | 4 | 7     | 9 | 9 |   |   |  |  |  |  |  |
| 9    | 8 | 7 | 4 | 3 | 2 |  | 11 |  | 0 | 5     | 5 | y | 8 | 9 |  |  |  |  |  |
|      |   |   |   | 9 | x |  | 12 |  | 5 | 4     | 8 | 9 |   |   |  |  |  |  |  |
|      |   | 8 | 8 | 8 | 1 |  | 13 |  | 1 | 1     | 5 | 6 | 7 |   |  |  |  |  |  |
|      |   |   |   |   | 7 |  | 14 |  | 4 | 7     | 8 |   |   |   |  |  |  |  |  |

Key (Boys)  
1 | 8 means 81 cm

Key (Girls)  
8 | 0 means 80 cm

- (a) The interquartile range of the heights for boys is 30 cm.

Find  $x$ .

Answer  $x = \dots\dots\dots$  [3]



- (b) The median height of the girls is 117.5 cm.

Find  $y$ .

*Answer*  $y = \dots\dots\dots$  [2]

- (c) Janet comments that on average, the girls are taller than the boys.

Explain why she is right.

*Answer*

[1]

- 10** A solid cylinder has radius  $2r$  cm and height  $h$  cm.  
A solid hemisphere has radius  $3r$  cm.  
The volume of the hemisphere is twice the volume of the cylinder.
- Work out, in terms of  $\pi$  and  $r$ , the total surface area of the cylinder.

*Answer* .....  $\text{cm}^2$  [4]

- 11** Julius is planning a trip from Singapore to Bintan.  
There are 3 daily ferries from Singapore to Bintan.  
On any particular day, the probability that a ferry departs on time is 0.85 .

**(a)** Find the probability that all 3 ferries on a particular day depart late.

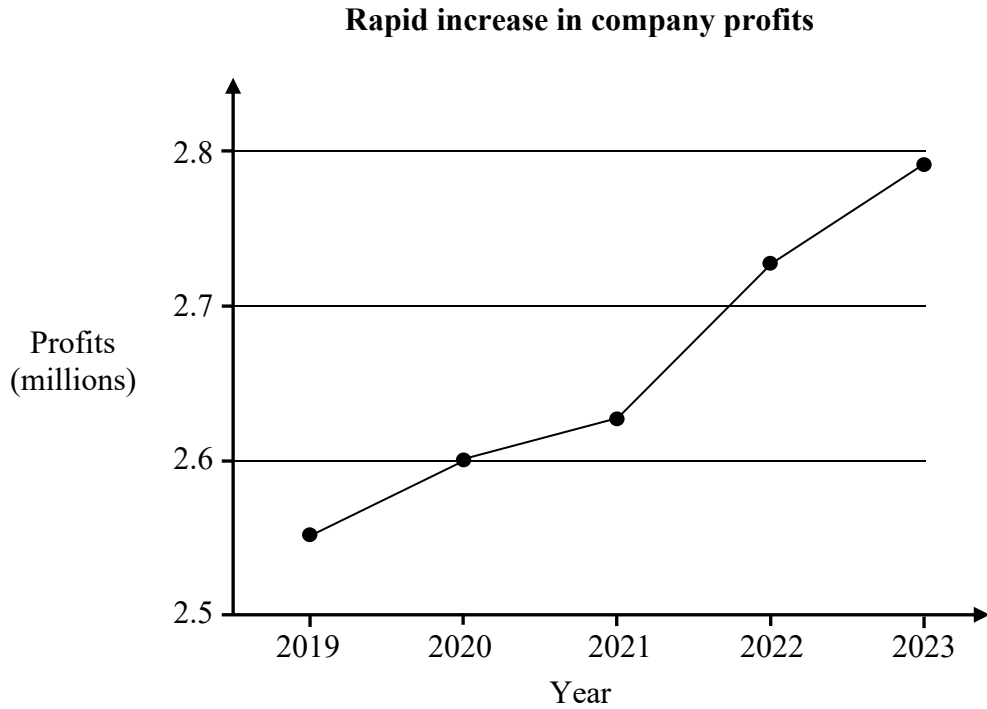
*Answer* ..... [2]

**(b)** Find the probability that at least one ferry departs on time on a particular day.

*Answer* ..... [1]

**(c)** Find the probability that exactly 2 out of the 3 ferries on a particular day depart on time.

*Answer* ..... [2]



- (a) State one misleading feature of the graph.

.....

..... [1]

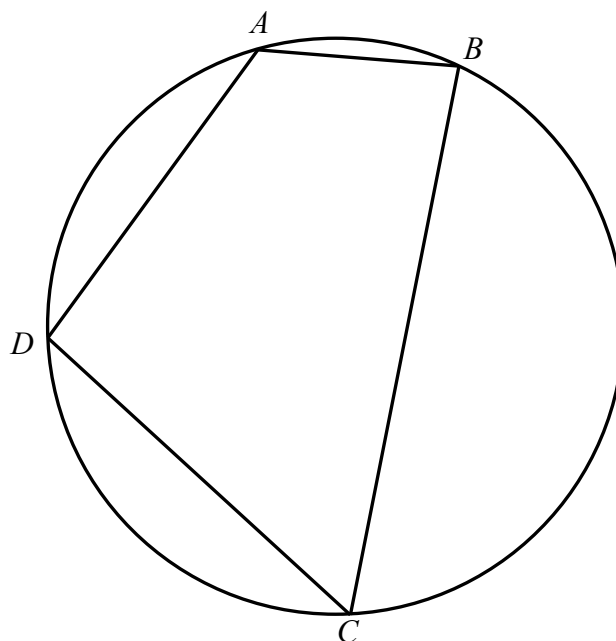
- (b) Explain how this feature affects the reader's interpretation of the graph.

.....

.....

..... [1]

- 13 The park is enclosed by a circular fence such that the vertices  $A$ ,  $B$ ,  $C$  and  $D$  of the park lie on the circumference of the fence.

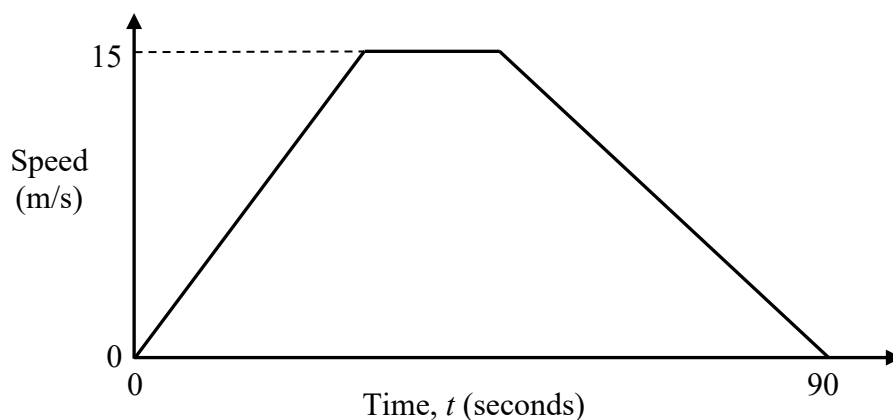


- (a) Construct the perpendicular bisector of  $BC$ . [1]
- (b) Construct the bisector of angle  $BCD$ . [1]
- (c) A toilet is to be constructed within the area enclosed by the fence but outside the park. The toilet is nearer to  $CD$  than to  $BC$  and nearer to  $C$  than to  $B$ .  
Mark a possible point,  $X$ , where the toilet is to be constructed. [1]

- 14  $p$  is directly proportional to  $q^3$ .  
It is known that  $p = 24$  for a particular value of  $q$ .  
Find the value of  $p$  when this value of  $q$  is halved.

*Answer*  $p = \dots\dots\dots$  [3]

15



The diagram shows the speed-time graph of a bus over a period of 90 seconds.

The bus reaches a maximum speed of 15 m/s.

- (a) Given that the acceleration was  $0.5 \text{ m/s}^2$ , calculate the time taken, in seconds, to reach its maximum speed.

*Answer* ..... s [1]

- (b) The total distance travelled during the 90 seconds was 750 m.  
Calculate the length of time that the bus was travelling at its maximum speed.

*Answer* ..... s [3]

- 16 (a) Express  $x^2 + 7x + 10$  in the form  $(x + p)^2 + q$ .

*Answer* ..... [2]

- (b) Write down the coordinates of the minimum point of the graph of  $y = x^2 + 7x + 10$ .

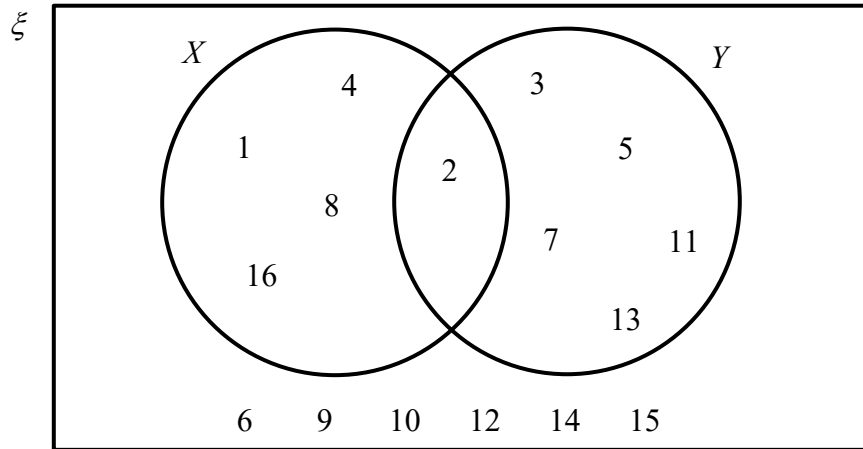
*Answer* ( ..... , ..... ) [1]

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- 17 (a)  $\xi = \{\text{integers } x : 1 \leq x \leq 16\}$   
 $X = \{1, 2, 4, 8, 16\}$   
 $Y = \{2, 3, 5, 7, 11, 13\}$

This information is shown on the Venn diagram.



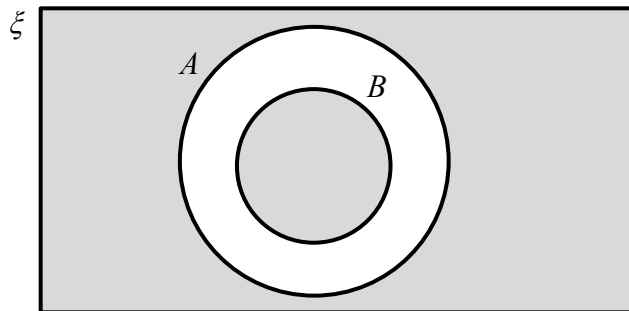
- (i) Describe the elements of set  $X$ .

.....  
 ..... [1]

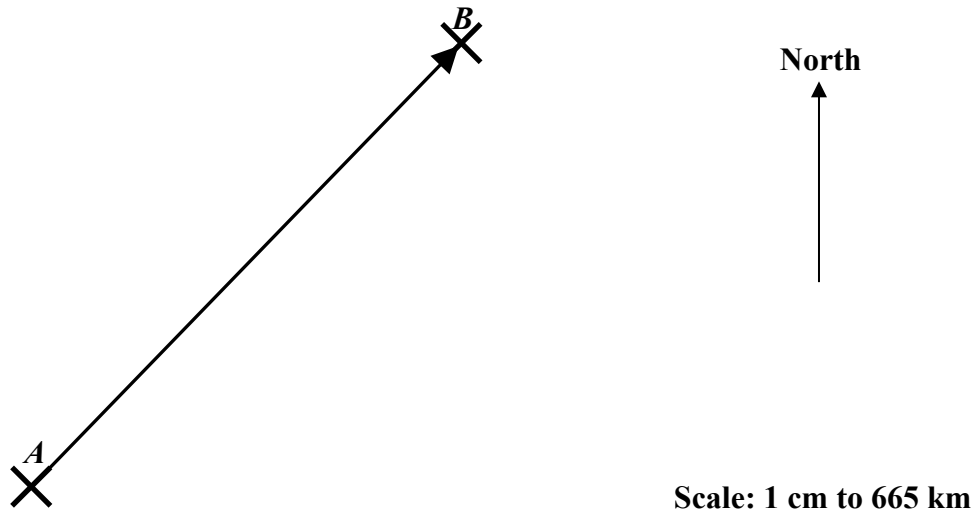
- (ii) List the elements in  $X' \cap Y$ .

Answer ..... [1]

- (b) Use set notation to describe the shaded region.



Answer ..... [1]



The diagram is a scale drawing showing the positions of two airports,  $A$  and  $B$ .  
The line  $AB$  shows the flight route of a plane flying from airport  $A$  to airport  $B$ .

- (a) Find the bearing of  $B$  from  $A$ .

*Answer* .....° [1]

- (b) The plane flies at an average speed of 730 km/h.

Find the time taken for the plane to travel from  $A$  to  $B$ .  
Give your answer in hours and minutes, correct to the nearest minute.

*Answer* ..... hours ..... minutes [3]

- 19 Each term in this sequence is found by adding the same number to the previous term.

$$p, \quad 15, \quad q, \quad r, \quad 39$$

- (a) Find the values of  $p$ ,  $q$  and  $r$ .

*Answer*  $p = \dots\dots\dots$

$q = \dots\dots\dots$

$r = \dots\dots\dots$  [2]

- (b) Write down an expression, in terms of  $n$ , for the  $n$ th term.

*Answer*  $\dots\dots\dots$  [1]

- (c) Explain why 3012 is not a term of this sequence.

*Answer*

.....  
 .....  
 ..... [1]

20  $P$  is the point  $(4, -3)$  and  $Q$  is the point  $(-9, 1)$ .

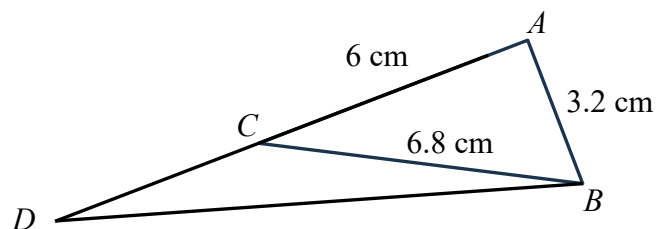
(a) Find the length of the line  $PQ$ .

*Answer* ..... units [2]

(b) Find the equation of the line  $PQ$ .

*Answer* ..... [2]

- 21 In triangle  $ABC$ ,  $AB = 3.2$  cm,  $AC = 6$  cm and  $BC = 6.8$  cm.



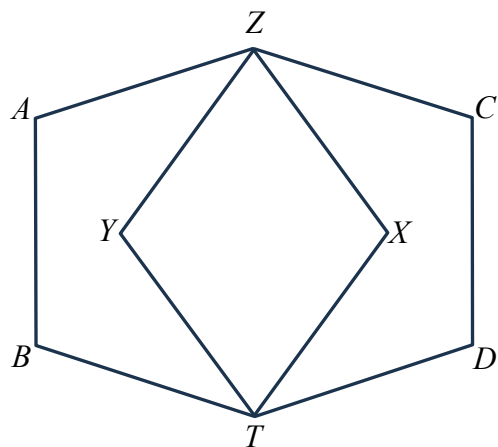
- (a) Show with mathematical working that  $AB$  is perpendicular to  $AC$ .

*Answer*

[2]

- (b) Find the exact value of  $\cos \angle BCD + \sin \angle ABC$ .  
Show your workings clearly.

*Answer* ..... [2]



The diagram shows two congruent regular pentagons,  $ZABTX$  and  $ZCDTY$ .

- (a) What is the special name given to the quadrilateral  $ZXTY$ ?

Answer ..... [1]

- (b) Find the interior angle  $ZXT$  of the pentagon  $ZABTX$ .

Answer ..... $^{\circ}$  [2]

- (c) Find the reflex angle  $ZYT$ .

Answer ..... $^{\circ}$  [1]

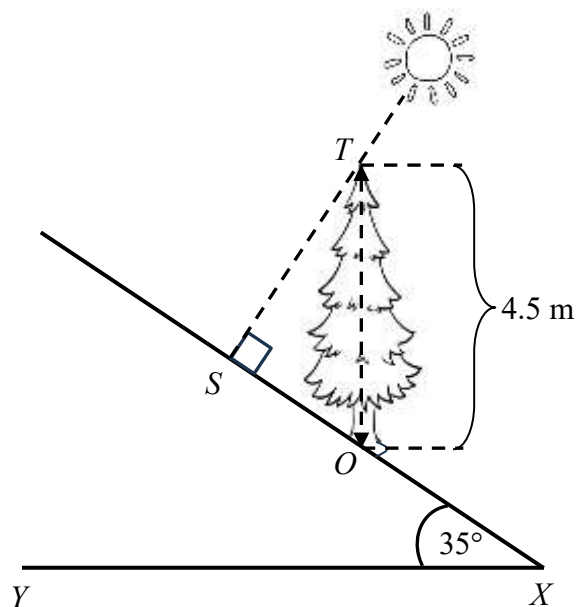
(d) Find the angle  $AZY$ .

*Answer* ..... $^{\circ}$  [2]

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- 23 The diagram shows a tree  $OT$  of height 4.5 m that stands vertically on a slope inclined at  $35^\circ$  with the horizontal  $XY$ .

At a particular time in the morning, the tree casts a shadow,  $OS$ , on the slope, such that  $TS$  is perpendicular to the slope.



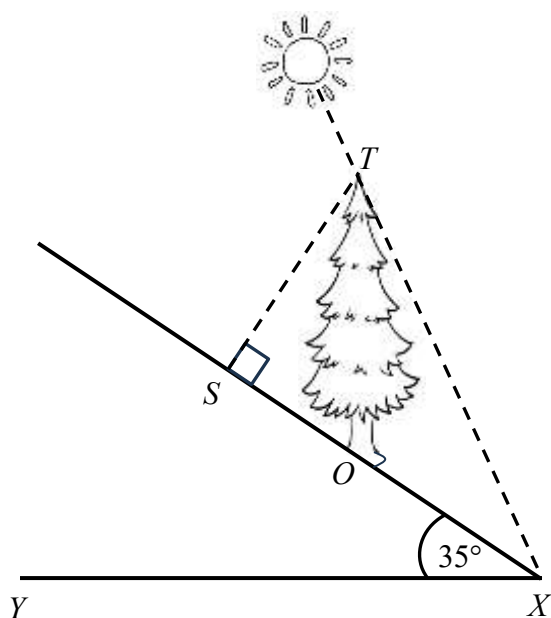
- (a) Calculate the length of the shadow  $OS$ .

Answer ..... m [2]



After some time, the sun changes its position as shown below.

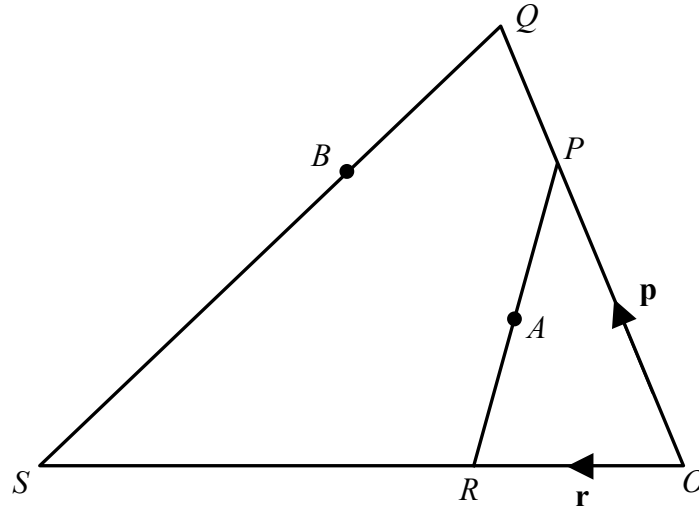
The tree then casts a shadow,  $OX$ , on the slope such that the length of this shadow is 3 m.



- (b) Find the angle that the sun makes with the horizontal  $XY$ .

Answer .....° [3]

24



In the diagram,  $\overrightarrow{OP} = \mathbf{p}$ ,  $\overrightarrow{OR} = \mathbf{r}$ ,  $\overrightarrow{OP} = 2\overrightarrow{PQ}$ ,  $\overrightarrow{RS} = 2\overrightarrow{OR}$  and  $\overrightarrow{RP} = 2\overrightarrow{RA}$ .

(a) Express, as simply as possible, in terms of  $\mathbf{p}$  and  $\mathbf{r}$ ,

(i)  $\overrightarrow{RP}$ ,

Answer ..... [1]

(ii)  $\overrightarrow{OA}$ ,

Answer ..... [1]

(iii)  $\overrightarrow{QS}$ .

Answer ..... [1]

(b)  $B$  is the point on  $QS$  such that  $QB : BS = 1 : 2$ .

(i) Express  $\overrightarrow{OB}$  in terms of  $\mathbf{p}$  and  $\mathbf{r}$ .

Answer ..... [2]

(ii) Hence write down two facts about  $O$ ,  $A$  and  $B$ .

.....

.....

..... [1]

(iii) Given that the area of triangle  $APO$  is  $12 \text{ cm}^2$ , find the area of triangle  $BOQ$ .

Answer .....  $\text{cm}^2$  [2]

**End of Paper**

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